



«Approved»
by the Dean
Syzdykova Z.A.
« » of 2024

Syllabus
Academic Year 2024-2025

1. General Information	
Course title	Mathematics for AI
Degree cycle (level)/major	6B04103 - AI Business
Year, term	1, 1
Number of credits	5
Language of delivery:	English
Prerequisites	Basics of programming and mathematics
Postrequisites	The knowledge obtained during the course will be used for the courses: • AI Business Fundamentals • AI for Medicine and Healthcare • AI for Production/Manufacturing • AI for Accounting Decisions • AI for Finance Decisions • AI for Marketing Decisions • AI for Logistics and Supply Chains • AI for HR Management • NLP, Generative AI in Business • AI Strategy for Competitive Advantage • Machine Learning for Business Applications • AI Ethics and Responsibility • Robotics for AI Business • Deep Learning for Business
Lecturer(s)	Meirbekova Bibinur, PhD, Assistant Professor, b.meirbekova@astanait.edu.kz , Astana IT University, Expo, office C1.1.337)
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	The course equips students with the mathematical foundations necessary for understanding and developing artificial intelligence (AI) algorithms and models. It covers key areas such as linear algebra, calculus, probability, statistics, and discrete mathematics, all tailored to AI applications. Through lectures, problem-solving sessions, and practical examples, students learn to apply these mathematical concepts to design and improve AI systems, including machine learning models, neural networks, and optimization algorithms. The curriculum emphasizes both theoretical understanding and practical skills,

	preparing students to tackle complex AI challenges with mathematical rigor and creativity.
2. Course Goal(s)	This course's goal is to provide students with essential mathematical knowledge and skills required for AI development. The course focuses on applying key mathematical concepts to AI algorithms and models, bridging theory with practical applications. Students will gain the ability to analyze, design, and optimize AI systems using a strong mathematical foundation.
3. Course Objectives:	• Develop a strong foundation in linear algebra, calculus, probability, and statistics relevant to AI. • Apply mathematical concepts to formulate and solve AI problems. • Understand the mathematical principles underlying common AI algorithms and models. • Implement mathematical methods for optimizing AI systems. • Analyze and interpret AI results using statistical and probabilistic techniques.
4. Skills & Competencies	On completion of the course, the student will be able to: • Perform vector and matrix operations essential for AI algorithms • Apply linear transformations and eigenvalue decomposition in machine learning contexts • Implement gradient descent and other optimization techniques for model training • Calculate probabilities • Analyze and interpret statistical measures in the context of machine learning datasets • Derive and manipulate probability distributions common in AI applications • Apply multivariate calculus techniques to optimize complex AI models • Implement numerical methods for solving equations and approximating functions in AI systems • Use information theoretic measures like entropy in decision tree algorithms • Mathematically describe and analyze the behavior of artificial neural networks • Formulate AI problems as mathematical optimization problems • Apply linear programming techniques to solve constrained optimization problems in AI • Use mathematical software tools (e.g., MATLAB, Python with NumPy) for AI-related computations • Interpret and critique the mathematical basis of current AI research papers
5. Course Learning Outcomes:	By the end of this course students will be able to: • Understand and apply linear algebra concepts crucial to AI algorithms

	• Demonstrate proficiency in probability theory and its applications in machine learning • Apply calculus techniques to optimize AI models • Utilize statistics for data analysis and model evaluation • Implement numerical methods relevant to AI computations • Comprehend information theory basics and their relevance to AI • Understand the mathematical foundations of neural networks • Utilize mathematical modeling techniques for AI applications • Demonstrate ability to translate AI problems into mathematical frameworks
6. Methods of Assessment	- Quizzes; - Theoretical and practical assignments;
7. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). <u>Basic Literature:</u> 1. C.L. David, R.L. Steven, J. M. Judi, Linear Algebra and Its Applications, Malaysia: Pearson Education Limited, 2019, 510p., ISBN 978-1-292-09223-2 2. L. Seymour, M.L. Lipson, Schaum's Outlines of Linear Algebra, 6 ed., USA: McGraw-Hill, 2018, 425p., ISBN 978-1-260-01144-9 3. Hala Nelson, Essential Math for AI. Next-Level Mathematics for Efficient and Successful AI Systems, O'Reilly Media, 2023 4. Tamoghna Ghosh, Shravan Kumar Belagal Math, Practical Mathematics for AI and Deep Learning. A Concise yet In-Depth Guide on Fundamentals of Computer Vision, NLP, Complex Deep Neural Networks and Machine Learning, first edition 2023, Copyright, BPB Publications, India, ISBN: 978-93-5551-194-2 <u>Supplementary literature:</u> 1. Lecture notes (available on moodle.astanait.edu.kz)
1. Resources	https://github.com/dair-ai/Mathematics-for-ML https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://github.com/wdp-007/Deep-learning-books/blob/master/3.%2520Math%2520Books/Mathematics%2520for%2520machine%2520learning-2017.pdf&ved=2ahUKEwjHmdjDlamHAxVDFRAIHVhiA4gQFnoECBYQAQ&usg=AOvVaw2PA0YHcJBvJn2vvteDUyJj

	https://github.com/Ian729/AI6104-MATHEMATICS-FOR-ARTIFICIAL-INTELLIGENCE https://github.com/quanghuy0497/Cheatsheet-collection
2. Course policy	Course and University policies include: Attendance: Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as “not graded”. In such case a student is not admitted to the exam and automatically fails the course. In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean’s office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more, he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing. Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. Class work: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student’s misconduct, the student can be dispensed from the classes. Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor. By the policy of this course, students who come

late to class for more than 5 minutes are now allowed to get in to class and consequently, they will be marked as “absent” for the specific hour.

Attestation I and II: Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course.

Home work / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone’s work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures.

Late submissions: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. Failure to pass assignment on time will result in 0% for the assignment. All gradings are based using a percentage grading scale.

In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.

Final exam: Final Quiz.

Laptops and mobile devices can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor.

Online lessons can be used in case if there won’t be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround.

Cheating and plagiarism are defined in the Academic conduct policies of the university and include:

1. Submitting work that is not your own papers, assignments, or exams;
2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation;
3. Submitting or using falsified data;
4. Submitting the same work for credit in two courses without prior consent of both instructors.

Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university.

	Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). Contacting the Course instructor: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously (on next workday). Instructor responds only on worktime: from Monday to Friday and from 9 am to 6pm. Instructor may not respond if request was answered once before or not related to education process, for example, request not planned by this course's syllabus additional tasks. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.
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3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practical sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	Algebra of Matrices: - Matrices - Matrix Operations (Addition, Scalar Multiplication, Multiplication, Transpose, Powers, The Inverse of a Matrix) - Rank of a Matrix	3	2	0	1	9
2	Determinant, Eigenvalue Problem, Matrix Diagonalization Problem: - Properties of Determinants - Characteristic Equation	3	2	0	1	9

3	Systems of Linear Equations: -Systems of linear equations, -Solving systems of linear equations (Row Operations, LU Decomposition), -Vectors and Linear Transformations,	3	2	0	1	9
4	Optimization: -Linear Programming (Geometric method, Simplex method) -Derivatives and Optimization -Gradients and Gradient Descent -Optimization in Neural Networks and Newton's Method	3	2	0	1	9
5	Probability and Statistics: - Introduction to Probability and Probability Distributions -Describing probability distributions and probability distributions with multiple variables -Sampling and point estimation -Confidence Intervals and Hypothesis testing	3	2	0	1	9
6	Probability and Statistics: - Introduction to Probability and Probability Distributions -Describing probability distributions and probability distributions with multiple variables -Sampling and point estimation -Confidence Intervals and Hypothesis testing	3	2	0	1	9
7	Differential Calculus and Gradients	3	2	0	1	9
8	Integral Calculus	3	2	0	1	9
9	Multivariable Calculus and Optimization	3	2	0	1	9
10	Neural Network: -Deep Learning, -Support Vector Machine (SVM)	3	2	0	1	9
	Total hours: 150	30	20	0	10	90

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	2	3	4	5
	Linear Algebra Fundamentals. Linear Programming	9	Reading list, books, internet resources Electronic version	Reading list, books, internet

				resources Electronic version
2	Vector Spaces and Transformations	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
3	Calculus Review	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
4	Probability Theory	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
5	Statistics for Machine Learning	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
6	Mathematical Foundations of Neural Networks	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1 st attestation	Assignments**: assignment 1, assignment 2, assignment 3 Mid Term Quiz Practical exam	60 20 20 20 40 16 24	100
2 nd attestation	Assignments**: assignment 4, assignment 5, assignment 6 End Term Quiz Practical exam	60 20 20 20 40 16 24	100
Final exam*	Quiz		100
Total	0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final		100

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent

A-	3,67	90-94	Good
B+	3,33	85-89	
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	
F	0	0-24	Fail

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught

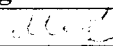
0-24	-No significant assessable material, absent or assessment missing a must pass component -The program code does not compile or the program does not work, i.e. does not read input or output correct result, or throw errors on execution. In such cases, a final grade of 0 is awarded.
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5. Methodological Guidelines

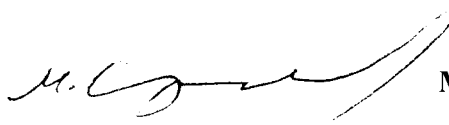
Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both quiz and written exam to evaluate the student's academic performance and professional skills.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Meirbekova Bibinur	Assistant Professor	26/08/24	

Director of Department of
Computational and Data Sciences



M.Zh. Sergaziyev